

# RAG Project Evaluation Rubric (Legal QA)

This rubric defines how your Retrieval-Augmented Generation (RAG) system will be evaluated across three scenarios.

| Scenario            | Required Metrics                                                                        | Scoring Focus                                                                      |
|---------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 1. Gold Q + A + Doc | Retrieval: Recall@k, MRR<br>Answer: EM/F1 or LLM Judge<br>Grounding: Faithfulness (LLM) | Full pipeline evaluation<br>Final = $0.35R + 0.4A + 0.25G$<br>(A,G penalized by R) |
| 2. Gold Q + A       | Answer: EM/F1 + LLM Judge<br>Semantic Similarity                                        | Answer quality evaluation<br>Final = $0.7A + 0.3Sim$                               |
| 3. No Gold Data     | Relevancy (LLM)<br>Faithfulness (LLM)<br>Coherence (LLM)                                | Proxy evaluation<br>Final = avg(scores)                                            |

## Key Principles

- Retrieval must be evaluated when gold documents exist.
- Correct answers without grounding are penalized.
- RAG systems are evaluated end-to-end, not just by answer quality.
- LLM-based evaluation is required when gold data is unavailable.